

Offline Parameter Estimation of Permanent Magnet Synchronous Machines by means of LS Optimization

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Abstract—Industrial applications, especially automotive ones should be robust and cheap. Both properties can be improved by using model based state estimation. Sensor cost can be reduced if some signal values are calculated from the other, already measured signals or the robustness of the system can be increased by supervising the sensors by calculating their measurement value out of the existing signal values. Robustness and redundancy is extremely important considering drive-by-wire technology, where the physical connection between the steering wheel and the wheels of the vehicle is omitted. This paper reports advances in permanent magnet synchronous machine model identification. By measuring machine input voltages, output currents speed and using the least squares optimization method, internal parameters of the machine can be estimated. In the identification stage, the model excitation signals are the current values and the speed of the machine and the response signals are the input voltages. After having a properly identified model, the output currents and electrical torque of the machine can be calculated knowing the input voltages and the speed of the machine. Those current sensors can be either eliminated or supervised by the model based redundant information.

Keywords: *permanent magnet synchronous machine, drive-by-wire, model identification, least squares method*

I. INTRODUCTION

Permanent Magnet Synchronous Machines (PMSM) are used in a wide variety of industrial applications. Because of their compact design and high efficiency, they are preferred in Electric Power Assisted Steering (EPAS) systems. Steering systems are safety-critical applications; therefore, they have to be designed by maximizing safety, redundancy and reliability. Automotive environment postulates further special requirements for EPAS applications, as e.g., the usage of unregulated and usually low voltage levels and providing high torque at high speeds with small size and high efficiency. The system cost also plays an important role in automobile system design because in high volume production every additional part may dramatically increase the overall cost. Designing cheap and reliable systems forces the engineers to build as few sensors as possible into the system and extract as much information as possible from the output of the sensors. One possible realization of this design principle is to implement redundant functions, which is required for safety checking,

without using redundant sensors, but rather using the available sensor data and model based simulations to calculate the appropriate value. In this paper, this technique is used in the following way: the machine input voltage signals and machine speed are measured, and using a machine model with properly identified parameters, the machine currents are estimated. The precision of the model-based estimation highly depends on the model parameters. Therefore, it is necessary to determine the model parameters precisely. In this paper, the authors are introducing the identification problem of PMSM and solving it using the least squares (LS) method [1]–[3]. The same problem is solved by the authors using other techniques as it can be seen in [4], [5]. Other authors have already presented parameter identification methods [6], [7] for PMSM drives, but these articles deal only with the identification of the mechanical parameters of the machine, considering the electrical parameters to be known and constant.

The main goal of this paper is to present a method to identify the electrical parameters. There were other publications on this topic [8] but, in that paper, the machine inductivity was neglected. That consideration results in a model which can be used only at slow machine speed. However electrical machines used in EPAS applications are driven at high speeds and have a nonnegligible inductivity parameter. In [9], an identification of stepper motor model was performed using LS method. Although [9] deals with a different machine type, that machine can be described using the same differential equations. In [9], the author used a machine model where the inductivity is not a function of the rotor angle. However, in many applications, it is essential to have rotor angle dependent inductivity and it is especially needed for reluctance variationbased sensorless rotor position detection algorithms [10]. Using this algorithm, it is necessary to have 10-30% difference between the lowest and the highest inductivity of the machine. In this paper, a more advanced motor model will be presented introducing rotor angle dependent inductivity. This model is described in a reference frame rotating synchronously with the rotor (d,q model) (2) and Fig. 1. The parameters of this motor model will be identified using LS method. PMSM model parameter estimation will be described in the following order:

First, the physical model of the machine will be expounded, then the simulation and the measurement environment will be detailed which are used to generate data for the identifications. After that the parameter estimation method will be described followed by the evaluation of the estimation results. Finally further research possibilities and a conclusion will be presented.

II. NOMENCLATURE

NOMENCLATURE

λ	Maschine's generator constant
ω_{el}	Electrical angular speed
ω_{mech}	Mechanical angular speed
Θ_r	Angle between rotor flux axis and phase u direction
$I_d(t), I_q(t)$	d,q axis current in rotating ref. frame
$I_d^*(t), I_q^*(t)$	Simulated d,q axis current
I_α, I_β	Currents in 2 axis steady reference frame
L_d	D axis inductance in rotating ref. frame
L_q	Q axis inductance in rotating ref. frame
L_u, L_v, L_w	Phase U,V,W inductance in 3 axis steady ref. frame
n_p	Number of rotor magnetic pole pairs
R_s	Stator resistance in rotating ref. frame
T_s	sample time
$U_d(t), U_q(t)$	d,q axis voltage in rotating ref. frame
ω_{el}	Electrical angular speed vector
ω_{mech}	Mechanical angular speed vector
e, e^2	Cost function values of the LS method
I_d^*, I_q^*	Simulated d,q axis current vector
I_d	Direct (d) axis current vector
I_q	Quadrature (q) axis current vector
P	Parameter vector of the LS method
$U[n]$	Output vector of the LS method
U_d, U_q	d,q axis voltage vector
$W[n]$	Regressor matrix of the LS method
EPAS	Electric Power Assisted Steering Systems
LS	Least Squares method
PMSM	Permanent Magnet Synchronous Machines
PWM	Pulse Width Modulation

III. PMSM MODEL

To estimate currents, a model derived from the physical model of the machine is used. This model takes the following electrical properties into account:

- resistance of the phase windings,
- voltage induced by the rotating magnetic field of the rotor magnets,
- the inductivity of the phase windings, and
- the coupling caused by phase windings.

The model does not treat the followings:

- eddy currents,
- resistance change of the phase windings caused by thermal effects,
- non-linear magnetic saturation and hysteresis effects of the iron parts and

- magnetic working point changes caused by the temperature change of the rotor magnets.

These effects can also be taken into consideration in a more complex model, but even with a simple model good results can be achieved. The model describes the PMSM assuming: resistance value of every phase coils are equal, the rotor magnets generate sinusoidal voltage signals, and the inductivity of the coils can be described using (1).

$$\begin{aligned} L_u(\Theta_r) &= \frac{L_d + L_q}{2} + \frac{L_q - L_d}{2} \cdot \cos(\Theta_r) \\ L_v(\Theta_r) &= L_u(\Theta_r + \frac{2\pi}{3}) \\ L_w(\Theta_r) &= L_u(\Theta_r + \frac{4\pi}{3}) \end{aligned} \quad (1)$$

All type of electrical machines can be modeled using different coordinate systems [11]. The most simple model is described in a coordinate system, where the two axes (d,q) are fixed to the magnetic axis of the rotor. Electrical model of the machine in this coordinate system can be seen in Fig. 1, where I_d, I_q are the currents, U_d, U_q are the voltage inputs, R_s is the phase resistance, L_d and L_q are the direct and quadrature inductivities and λ is the generator constant of the machine [11]. ω_{el} is the electrical angular speed of the machine. ω_{el} can be calculated with the following formula: $\omega_{el} = n_p \cdot \omega_{mech}$, where ω_{mech} is the rotor mechanical speed and n_p is the number of rotor magnetic pole pairs [11]. Quadrature (q) current component is perpendicular to the rotor's magnetic field; it is used to generate an effective torque. Direct (d) current component is parallel to the rotor's magnetic field and is used to decrease the effect of the rotor's magnetic field on the stator windings (field weakening) [11]. In this model, the current and the voltage signals are not sinusoidal even if the rotor is rotating with a constant speed because the coordinate system of this model rotates synchronously with the rotor. In one mechanical working point, where the torque and the rotor speed are constant, all the currents and the input voltages are constant signals. It is also possible to describe the machine in the coordinate system fixed to the stator windings. This model is closer to the physical model of the machine. In this model, voltage and current signals are sinusoidal if the rotor is rotating [11]. The two machine models are equivalent. The authors prefer the (d,q) model because it is easier to understand the meaning of the constant signals. To switch between different machine models, Clarke and Park transformations are used. Describing transformations are out of scope of this paper. More detailed description about machine models and transformations can be found in [4], [11].

As it can be seen in Fig. 1 and in (2), the model describes the machine currents I_d, I_q by using the voltage inputs U_d, U_q in a rotor-oriented coordinate system.

$$\begin{aligned} U_d(t) &= R_s \cdot I_d(t) + L_d \cdot \frac{dI_d(t)}{dt} - L_q \cdot \omega_{el}(t) \cdot I_q(t) \\ U_q(t) &= R_s \cdot I_q(t) + L_q \cdot \frac{dI_q(t)}{dt} + L_d \cdot \omega_{el}(t) \cdot I_d(t) \\ &\quad + \omega_{el}(t) \cdot \lambda \end{aligned}$$

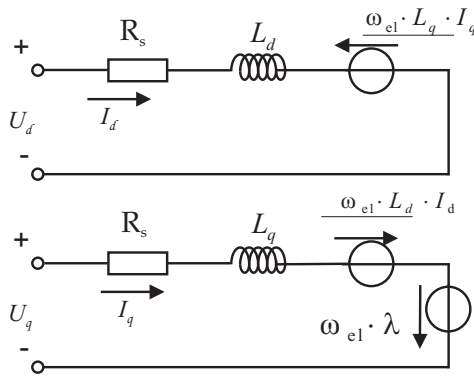


Figure 1. Rotor oriented reference frame

IV. SIMULATED SIGNALS

To identify system parameters, measurements are required. In this section a model-based simulation is detailed which generates measurement-like data to test the parameter estimation algorithm. The sampled time version (3) of PMSM model (2) introduced in section III was used in the simulation. All time-dependent variables ($x(t)$) was replaced by a sampled time variable ($x[n] = x(n \cdot T_s), \forall t \in [n \cdot T_s, (n + 1)T_s)$, where T_s is the sampling time). The voltage inputs of PMSM model was calculated using a motor control software model. During the simulation the PMSM input voltages, output currents and rotor speed was recorded. The simulation model can be seen in Fig. 2. The PMSM model and the motor control program was implemented using Simulink modelling tool of the MATLAB [12] computing program.

$$\begin{aligned}
U_d[n] &= R_s \cdot I_d[n] + L_d \cdot \frac{I_d[n] - I_d[n-1]}{T_s} \\
&\quad - L_q \cdot \omega_{el}[n] \cdot I_q[n] \\
U_q[n] &= R_s \cdot I_q[n] + L_q \cdot \frac{I_q[n] - I_q[n-1]}{T_s} \\
&\quad + L_d \cdot \omega_{el}[n] \cdot I_d[n] + \omega_{el}[n] \cdot \lambda
\end{aligned} \tag{3}$$

The following requirements are important for both simulation and measurement excitation signals. It is important to excite the system considering the following:

- this system has to be excited enough to have a good signal-to-noise ratio,
- different working points (mechanical speed and torque) should be used to have relevant information about the system behaviour in the whole working range of the machine.

Some special requirements should also be considered regarding the generation of simulated excitation signals: they should be a simple signal which can be reproduced during measurements so the identification results of simulations and measurements can be compared. The following speed (Fig. 3)

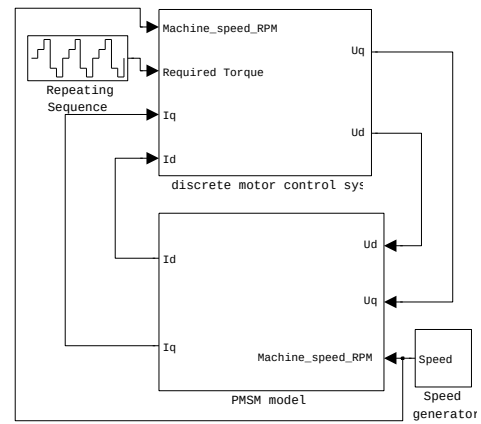


Figure 2. Simulation model

and required torque (Fig. 4) was used in simulation. Using this input signals, a motor control system generates the input signals for the PMSM model. During the simulation U_d (Fig. 5), U_q (Fig. 6), I_d (Fig. 7), and I_q (Fig. 8) was recorded.

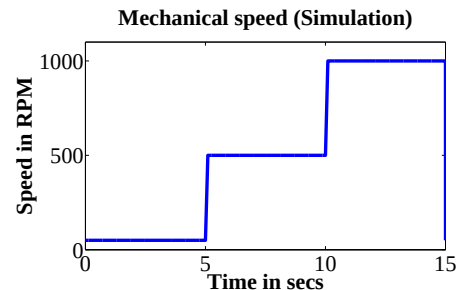


Figure 3. Reference machine speed in simulation

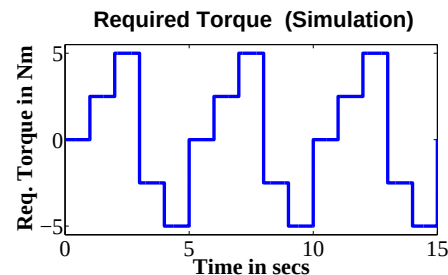


Figure 4. Required torque in simulation

V. MEASUREMENT ON REAL PMSM

Measurements were done to test the quality of the LS parameter estimation method in a real physical environment. Measurement configuration is shown in Fig. 9. The PMSM was driven by a load machine which was set up to rotate with the speed reference signal of the measurement. The motor control system was implemented in an Autobox rapid prototyping board. This embedded system was controlled by a host PC. The captured signals was stored on the hard drive

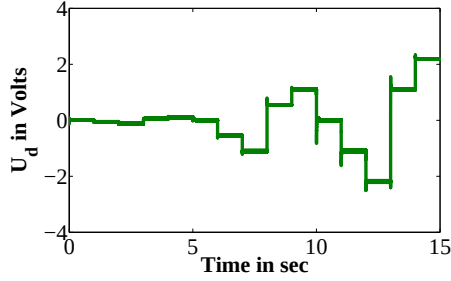


Figure 5. Simulated U_d

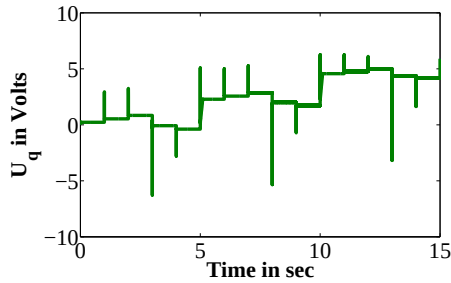


Figure 6. Simulated U_q

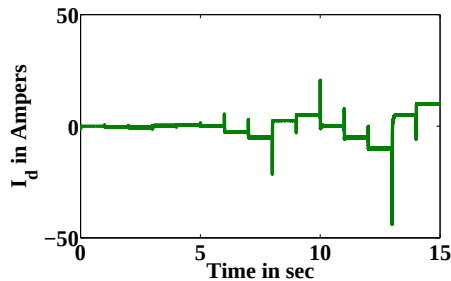


Figure 7. Simulated I_d

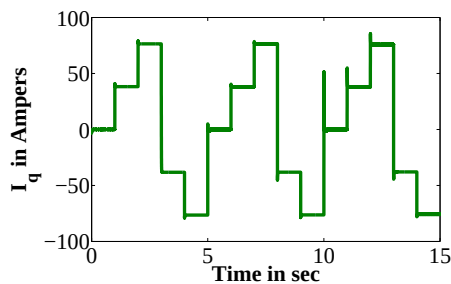


Figure 8. Simulated I_q

of the PC. The pulse width modulation (PWM) signals – generated by the Autobox – are processed by an inverter, which applies the voltage signals to the PMSM. The following

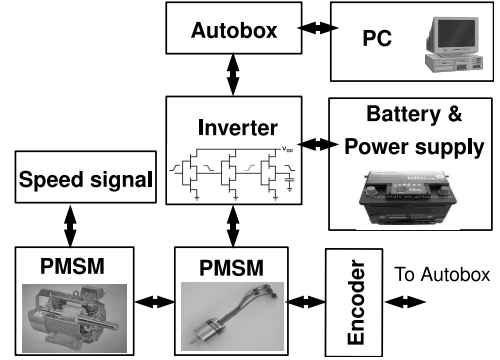


Figure 9. Measurement configuration

speed (Fig. 10) and required torque (Fig. 11) was used during measurement. The following signals U_d (Fig. 12), U_q (Fig. 13), I_d (Fig. 14), and I_q (Fig. 15) were recorded.

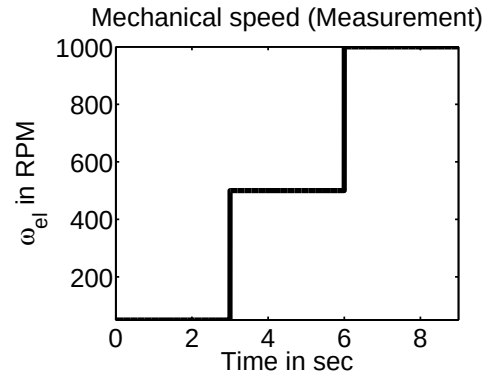


Figure 10. Reference machine speed during measurement

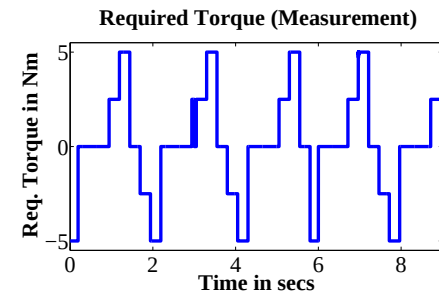


Figure 11. Required torque during measurement

VI. MODEL PARAMETER ESTIMATION

The goal of the optimization was to minimize the difference between the model-based ($I_d^*(t), I_q^*(t)$) and the measured ($I_d(t), I_q(t)$) machine currents. Unfortunately, the currents can not be written using a linear combination of the parameters, so the LS method can not be used directly to optimise an

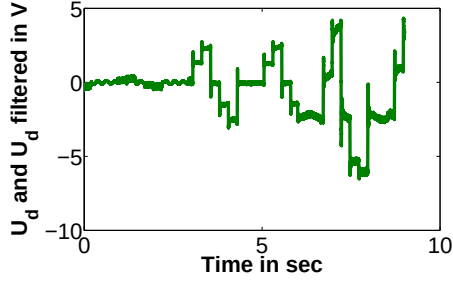


Figure 12. Recorded U_d

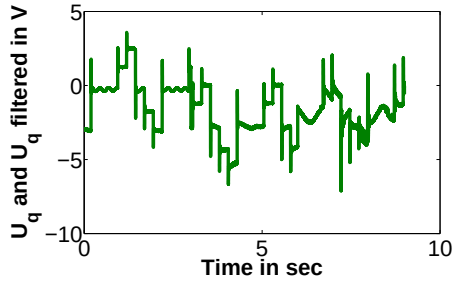


Figure 13. Recorded U_q

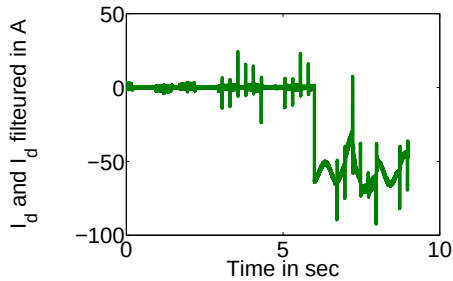


Figure 14. Measured I_d

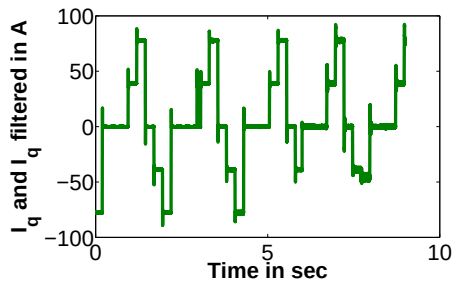


Figure 15. Measured I_q

error function such as (4), where the $\cdot^2$ sign means element-wise operation. Instead of the currents, the voltages can be expressed using a linear equation system as it is written in (5). PMSM model was identified in the following way: the input signals were the phase currents described in the rotating reference frame (I_d, I_q). The output signals were the voltages of the machine also described in rotating reference frame (U_d, U_q). The following machine parameters were identified:

- phase resistance (R_s), magnetic constant (λ),
- direct inductivity (L_d), quadrature inductivity (L_q).

Using simulation or measurement generated data vectors, the scalar values of the machine parameters were estimated. In the following sections, it will be verified that the LS method works well using (5). Also, weighted LS method can be used in case of bad signal-to-noise ratio to increase the estimation accuracy, but in this case the simple LS method also presented good results.

$$\mathbf{e}^2 = (\mathbf{I}_d^* - \mathbf{I}_d)^2 + (\mathbf{I}_q^* - \mathbf{I}_q)^2 \quad (4)$$

$$\mathbf{e}^2 = (\mathbf{U}_d^* - \mathbf{U}_d)^2 + (\mathbf{U}_q^* - \mathbf{U}_q)^2 \quad (5)$$

The improved version of machine model described in [9] is (6)

$$\mathbf{U}^*[n] = \mathbf{W}^T[n] \mathbf{P}, \quad (6)$$

where $\mathbf{U}^*[n]$ given by (7) is the estimated phase voltage vector of the machine, $\mathbf{W}^T[n]$ given by (8) is the regressor matrix, $\mathbf{P}$ given by (9) is the vector of machine parameters and operator $\cdot^T$ means vector or matrix transpose.

$$\mathbf{U}[n] = \begin{bmatrix} U_d[n] \\ U_q[n] \end{bmatrix}, \quad \mathbf{U}^*[n] = \begin{bmatrix} U_d^*[n] \\ U_q^*[n] \end{bmatrix} \quad (7)$$

$$\mathbf{W}[n] = \begin{bmatrix} \frac{I_d[n] - I_d[n-1]}{T_s} & I_q[n] \\ -\omega_{el}[n] \cdot I_q[n] & \frac{I_q[n] - I_q[n-1]}{T_s} \\ 0 & \omega_{el}[n] \end{bmatrix} \quad (8)$$

$$\mathbf{P} = \begin{bmatrix} R_s \\ L_d \\ L_q \\ \lambda \end{bmatrix} \quad (9)$$

Now the the error (10) has to be minimised. The parameter set $\mathbf{P}$, which minimises the error according to the LS method [2] is defined by (11), where n_{max} is the last sample of the simulation or measurement data.

$$\mathbf{e} = \sum_{n=2}^{n_{max}} \mathbf{W}^T[n] \cdot \mathbf{P} - \mathbf{U}[n] \quad (10)$$

$$\mathbf{P} = \left(\sum_{n=2}^{n_{max}} \mathbf{W}^T[n] \cdot \mathbf{W}[n] \right)^{-1} \left(\sum_{n=2}^{n_{max}} \mathbf{W}[n] \cdot \mathbf{U}[n] \right) \quad (11)$$

Table I
REAL AND ESTIMATED VALUES OF PARAMETERS

	R_s in Ω	L_d in μH	L_q in μH	λ in A/sec
Real val.	0.037500	115.00	125.00	0.010500
Est. val. LS method	0.037500	115.00	124.99	0.010500

VII. ESTIMATION RESULTS OF SIMULATED INPUT SIGNALS

In this section, the quality of the LS parameter estimation is evaluated using data files generated by simulation model. To validate the parameter estimation algorithm, numerous estimations are made using different data and random initial parameters. In Table I, the estimated parameter values using LS method can be compared with the real values used for generating the simulation data.

VIII. ESTIMATION RESULTS OF MEASURED INPUT SIGNALS

In this section, the quality of the LS parameter estimation is evaluated using measurement data files. After successful parameter estimation, the machine currents were estimated using the machine model. The measured and the estimated currents can be seen in Fig. 16 and in Fig. 17. The figures demonstrate that the estimation method performs well.

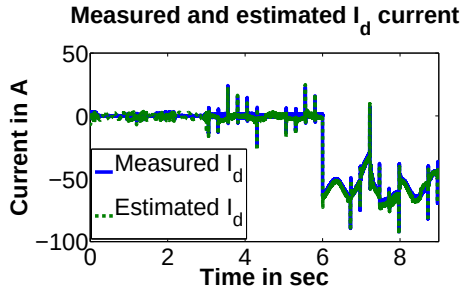


Figure 16. Measured and estimated I_d currents

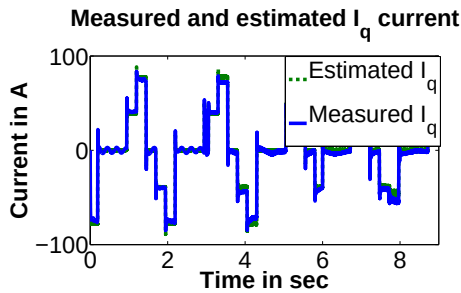


Figure 17. Measured and estimated I_q currents

IX. CONCLUSION AND FURTHER RESEARCH PLAN

The main goal of the research was to identify a PMSM model. If the model based current estimations are reliable enough to replace the current sensor information, then the

machine can be controlled and machine torque can be estimated without properly working current measurement sensors. A model has been presented which resulted in good current estimation. In this paper, an LS method was shown, which is suitable for offline machine parameter estimation. The presented method works well both on simulation and on measurement collected data. For further development an on-line, real time machine parameter estimation can be investigated. Using on-line identification motor health monitoring can be implemented.

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